Discrete Mathematics

Unit 1: Sets, Relations and Functions

A set is an unordered collection of distinct objects. For sets A and B: A ∪ B, A ∩ B, A − B and the complement A'. De Morgan's laws: (A ∪ B)' = A' ∩ B' and (A ∩ B)' = A' ∪ B'. |P(A)| = 2^|A|.

Relations

A relation R on A is a subset of A × A. R is reflexive, symmetric, antisymmetric or transitive. An equivalence relation (reflexive, symmetric, transitive) partitions A into equivalence classes. A partial order (reflexive, antisymmetric, transitive) can be drawn as a Hasse diagram.

Functions

f: A → B is injective if f(a1) = f(a2) implies a1 = a2, surjective if every b has a preimage, and bijective if both. Composition: (g ∘ f)(x) = g(f(x)).